

Jakob von Uexküll

Ton (“tone”), Stimmung (“mood”), and the Tonal Constitution of the Umwelt (“surrounding-world,” “phenomenal world,” “self-world,” “environment”)

A Detailed Analysis of A Foray into the Worlds of Animals and Humans (1934) and A Theory of Meaning (Bedeutungslehre, 1940)

Source: Jakob von Uexküll, *A Foray into the Worlds of Animals and Humans, with A Theory of Meaning*, trans. Joseph D. O’Neil (University of Minnesota Press, 2010). Original German titles: *Streifzüge durch die Umwelten von Tieren und Menschen* (“*Forays through the surrounding-worlds of animals and humans*,” 1934) and *Bedeutungslehre* (“*Theory of meaning*,” 1940). Primary chapters under interpretation: Foray Chapter 1 (“Foreword,” pp. 41–43) and Chapter 2 (“Introduction,” pp. 44–52); Chapter 9 (“Perception Image and Effect Image,” pp. 92–97); Chapter 13 (“Search Image and Search Tone,” pp. 113–118); Chapter 14 (“Magical Environments,” pp. 119–125); Chapter 15 (“The Same Subject as Object in Different Environments,” pp. 126–132); *Bedeutungslehre* Chapter 1 (“Carriers of Meaning,” pp. 139–145); Chapter 7 (“The Composition Theory of Nature,” pp. 171–181); Chapter 12 (“Summary and Conclusion,” pp. 200–208). Page numbers refer to the published English edition.

1. The Problem and the Stakes

The aim of Uexküll’s biological-philosophical project, undertaken from his Hamburg laboratory between the wars and given its mature form in the 1934 *Foray* and the 1940 *Bedeutungslehre* (“Theory of meaning”), is to wrest the analysis of animal life out of two converging traditions that, on his diagnosis, have “lost the phenomenon” of the living being. The first is the *Maschinentheorie* (“machine theory”)—the post-Cartesian, post-Darwinian biology that treats organisms as objects governed by efferent and afferent reflex arcs, animated by mechanical causation, and selected by environmental pressure. The second is the equally obstinate *Vorstellungspsychologie* (“psychology of representations”)—the doctrine that an animal’s relation to its world is constituted by inner images that mirror an outer object-world held to be the same for all observers. Against both, Uexküll’s central thesis is that the living being is a *Subjekt* (“subject”) whose world is constituted, not surveyed; that this constituted world—the *Umwelt* (“surrounding-world,” “phenomenal world,” “self-world,” “environment”)—is

closed, complete, and saturated with what he will call *Töne* (“tones”); and that the living being’s inhabitation of its *Umwelt* is best understood not on the model of a machine receiving inputs, nor on the model of a mind picturing externals, but on the model of an instrument joining a melody.

The lectures of Heidegger that occupy the same cultural moment (the 1924 Marburg course, the 1927 *Being and Time*, the 1929/30 *Fundamental Concepts of Metaphysics*) circle the same problem from a different angle, with different stakes; the convergence between Uexküll and Heidegger is not coincidental but reflects a shared preoccupation in the German-speaking philosophical-biological-phenomenological milieu of the 1920s and 1930s with the question of how to wrest the analysis of finite living being from a Cartesian inheritance that had everywhere petrified the phenomenon. Heidegger explicitly cites Uexküll in *Sein und Zeit* §§14–18 and develops the *Umwelt* analysis from within Uexküll’s biological vocabulary; Uexküll’s *Stimmung* (“mood”)—as it appears in the hermit crab analysis of *Foray* Chapter 9—anticipates by several years what Heidegger will name *Befindlichkeit* (“state-of-mind,” “disposedness,” “finding-oneself”) in *Being and Time* §29. The relations are textual, not merely thematic; the biology and the phenomenology speak to each other in a shared idiom whose grammar is the grammar of *tonality*.

The analysis that follows takes up Uexküll’s theory of *Töne* and *Umwelt* through six concentric framings, each of which Uexküll develops at a different scale. *First*, the discovery of the *Umwelt* and the architecture of the *Funktionskreis* (“functional cycle”): the closed loop through which subject and world are mutually constituted (*Foray* Chapters 1–2). *Second*, the formal introduction of *Wahrnehmungsbild* (“perception image”) and *Wirkbild* (“effect image”), and the birth of *Wirkton* (“effect tone”) in the hermit-crab analysis (Chapter 9). *Third*, *Suchbild* (“search image”) and *Suchton* (“search tone”)—innate and acquired tonal templates that guide perceptual orientation prior to any specific stimulus (Chapter 13). *Fourth*, *magische Umwelten* (“magical environments”)—worlds in which tones generate phenomena without external trigger (Chapter 14). *Fifth*, the *Same Subject as Object in Different Environments* paradigm, where the oak tree appears as a different *Bedeutungsträger* (“carrier of meaning”) to forester, girl, fox, owl, squirrel, songbird, ant, bark beetle, and ichneumon wasp—each cut into the oak by a different tonal articulation (Chapter 15). *Sixth*, the *Bedeutungslehre*’s extension of *Töne* into the *Eigenton* (“self-tone”) of the living cell, the *Bildungsmelodie* (“formative melody”) of the developing organism, and the *Kompositionslehre der Natur* (“composition theory of nature”) in which all *Töne* sound together as a *Partitur* (“score”) that, on Uexküll’s late hyperbolic formulation, is “beyond space and time” (*Bedeutungslehre* Chapter 12). Throughout, the question is the same: what is a *tone*, on Uexküll’s account, and how does it constitute the *Umwelt* of a finite living being?

2. The Discovery of the *Umwelt* and the Architecture of the *Funktionskreis* (Foray Chapters 1–2)

2.1 The Anti-Mechanist Foreword

The Foreword opens with a polemical declaration whose terms remain decisive for the entire project: “Such excursions [into the worlds of animals] are still considered fanciful by many natural scientists who cling to the doctrine that all living beings are mere machines. Were they to be machines, they would be without an environment [*Umwelt*]” (p. 41). The first move is therefore not a definition but a *contrast*: the *Umwelt* is what a *Subjekt* has and a machine lacks. The grammar is decisive—the term *Umwelt* is not interchangeable with *Umgebung* (“surroundings”); the latter is the objective field that remains when one strips away the perceiving subject; the former is the closed perceptual-effective world that *only* a subject can have. Uexküll glosses the contrast through a metaphor he will return to throughout: each animal “is enveloped in a soap bubble [*Seifenblase*] which encloses its environment” (Foray, Chapter 1, paraphrased; cf. p. 43). The image registers two facts simultaneously: each *Umwelt* is bounded (the bubble is closed); each *Umwelt* is its own, not a mere section of a single shared world (every bubble is its own bubble).

The Foreword’s structural argument is that the *Umwelt* is composed of two coordinated dimensions: the *Merkwelt* (“perception world”), comprising everything the subject can perceive, and the *Wirkwelt* (“effect world”), comprising everything the subject can affect. Together they form a closed totality: “We no longer regard animals as mere objects, but as subjects whose essential activity consists in perceiving [*merken*] and effecting [*wirken*]. We thereby unlock the gates that lead to other realms.... All that a subject perceives becomes its perception world [*Merkwelt*] and all that it effects, its effect world [*Wirkwelt*]. Perception and effect worlds together form a closed unit, the environment [*Umwelt*]” (Foreword, p. 42). The Cartesian division between *res cogitans* (the inner mind picturing) and *res extensa* (the outer world surveyed) is replaced from the start by a different topology: a *Subjekt* is the closure of a *Merkwelt* + *Wirkwelt*, and the *Umwelt* is just this closure.

The Foreword introduces a further, easily-missed terminological move that will prove decisive for the tonal analysis. The standard German pair *Werkzeug* (“tool,” “instrument for effecting”) names what an organism uses to act on its world. Uexküll coins, alongside it, *Merkzeug* (“tool for perceiving,” “perception-tool”)—a neologism that bears the same morphology and grammatical force as *Werkzeug* but applied to the perceiving side. The pair *Merkzeug* / *Werkzeug* indexes a methodological symmetry: perception is not a passive registration but an active operation requiring its own instrumentation, just as effecting is. The animal *makes* its perception, just as it *makes* its effect; both are activities of a *Subjekt* in its *Umwelt*, not properties of an object in a world.

2.2 The Tick Paradigm

The Introduction opens with what is now the most famous biological *exemplum* in twentieth-century philosophy of life: the tick (*Ixodes ricinus*). Uexküll's analysis of the tick is structurally essential because it shows, in a being whose simplicity makes the tonal architecture visible, that an *Umwelt* can be both impoverished and complete—that completeness is a function of the closure of the *Funktionskreis* (“functional cycle”), not of the number of perceivable features. The tick lives by *three* perception-marks (*drei Merkmale*): butyric acid (signaling the proximity of a mammal), warmth (signaling skin contact), and the tactile quality of skin-with-hair (signaling the right place to drill). It executes *three* corresponding *Wirkmale* (“effect marks”): drop, attach, drill. The tick has waited for as long as eighteen years between life stages, the perception apparatus stilled until butyric acid arrives. “The tick can wait eighteen years. None of us can hold out that long” (p. 47).

The tick's *Umwelt* is, by any objective comparison, one of the poorest known. It is also—and this is Uexküll's polemical point—structurally identical to the human *Umwelt* in its fundamental ontology: closed *Funktionskreis*, perception apparatus geared to specific *Merkmale*, effector apparatus geared to specific *Wirkmale*, *Subjekt* constituted in the closure. The conclusion is rendered with full force: “All animal subjects, from the simplest to the most complex, are inserted into their environments to the same degree of perfection” (p. 50). The thesis subverts every evolutionary or developmental hierarchy that would rank the tick beneath the human; it puts forward instead a doctrine of *equal completeness through different closures*. Each *Subjekt* has its world, and its world is whole.

2.3 The *Funktionskreis* and the Anatomy of the Tonal Loop

The *Funktionskreis* is the structural unit through which the *Subjekt-Umwelt* relation is articulated. It is a closed loop, diagrammed by Uexküll in Figure 3 of the Introduction, with the following moments: (1) the object's *Merkmalträger* (“perception-mark carrier”) is registered by the subject's *Merkorgan* (“perception organ,” the brain organ grouping perception cells), generating a *Merkzeichen* (“perception sign”) inside the subject; (2) the *Merkzeichen* is “transposed” outward into the *Umwelt* as a *Merkmal* (“perception mark”), a quality the object now has *for the subject*; (3) the subject's response is initiated by its *Wirkorgan* (“effect organ”), generating a *Wirkzeichen* (“effect sign”) inside the subject; (4) the *Wirkzeichen* is realized externally as a *Wirkmal* (“effect mark”) imprinted on the object's *Wirkmalträger* (“effect-mark carrier”); (5) the *Wirkmal* “extinguishes” the *Merkmal*—the perception that triggered the action is consumed in the action's completion; (6) the cycle closes, and the subject is restored to readiness for a new initiation.

Three structural features of this schema bear on the tonal analysis. *First*, the object that figures in the *Funktionskreis* is not the “object-in-itself” of objectivist biology but the *Gegengefüge* (“counterstructure”)—the object's structural connection between its *Merkmalträger* and its *Wirkmalträger* as these are made available *to this subject's particular cycle*. The same object can enter multiple *Funktionskre-*

ise and be different in each; the tick's mammal-as-butyric-acid-source and the gardener's mammal-as-cow-to-be-milked are not the same *Gegengefüge* even when they are the same animal in the same field. *Second*, the *Merkzeichen* / *Wirkzeichen* internal to the subject is the locus where Uexküll grounds his neo-Kantian commitment: following Johannes Müller's principle of *spezifische Sinnesenergien* ("specific sensory energies"), every external stimulus that reaches a perception cell produces *that cell's particular sign*, regardless of the stimulus's external nature; what the subject perceives is therefore always its own internal sign, projected outward into the *Umwelt* as a *Merkmal*. The *Umwelt* is genuinely the subject's *self-world*. *Third*—and this is the seed of the tonal doctrine—the *Funktionskreis* is structured *by what action the perception is for*: the same external stimulus, processed through different functional cycles, generates different perception marks because it is held, by the subject's organic structure, to different effect marks. The *Wirkmal*—the imminent effect—colors the *Merkmal* before the *Merkmal* is fully constituted. This is the structural seed of the *Wirkton* ("effect tone").

2.4 The Kantian Frame

The Introduction concludes with two assertions whose philosophical weight is full Kantian: "Without a living subject, there can be no time" (p. 48), and—by extension—no space. Time is *Subjekt-relativ*: each species has its own *Moment* (Uexküll's technical term for the species-specific temporal quantum below which no event can be perceived as differentiated; the human *Moment* is approximately 1/18 of a second). Space, similarly, is constructed from *Richtungsschritte* ("directional steps") whose magnitude is species-specific (approximately 2 cm for the human). The space and time of an *Umwelt* are not the homogeneous coordinates of Newtonian physics; they are the *Anschaungsformen* ("forms of intuition") of a particular *Subjekt*, in Kant's idiom, given material biological articulation. "Biology connects with Kant's epistemology," Uexküll says—not as a piety but as a structural claim about the place of subjective constitution in the very being of the living being.

The Kantian commitment has consequences for the entire tonal analysis. If space and time are subject-relative, then the field within which *Töne* sound is itself constituted by the same subject whose *Töne* sound in it. There is no objective auditorium in which all *Umwelten* can be heard; each *Umwelt* is its own auditorium, with its own acoustics, and the *Töne* that fill it are audible *only* within it. This will become decisive when we reach the oak paradigm: the squirrel hears the *Klettertönung* ("climbing tone") of the bark, the bark beetle hears the *Fressston* ("feeding tone") of its underside, the ichneumon wasp hears the wood as soft-as-butter—and none of these tones is heard by any other inhabitant.

3. *Wahrnehmungsbild, Wirkbild, and the Birth of Ton (Foray Chapter 9)*

3.1 The Hermit Crab Experiment: The Paradigmatic Mood-Determined Tone

Chapter 9, “Perception Image and Effect Image,” is the philosophical center of the tonal analysis. Uexküll there reports a sequence of experiments on the hermit crab (*Eupagurus*) and the sea anemone (*Sagartia*) that, in his own retelling, do more philosophical work than any single passage in the *Foray*. The setup: a hermit crab in its snail-shell encounters a sea anemone. *In its normal state*, the crab grasps the anemone and places it on the shell, so that its stinging tentacles defend the soft abdomen against squid-attack. *In the same crab, with its shell removed*, the crab does not place the anemone on the shell—it tries instead to *crawl into* the anemone, using its body as a substitute shell. *In a starving crab*, the same anemone, the same sensory presentation, becomes prey: the crab eats it.

Uexküll’s interpretation: the *Wahrnehmungsbild* (“perception image”; literally “noticing-image”)—the integrated sensory representation of the anemone delivered by the crab’s perceptual apparatus—is identical in all three cases. What differs is the *Stimmung* (“mood,” “attunement”)—the internal disposition of the subject—and the *Stimmung* determines which of three possible *Wirkbilder* (“effect images”; “the representation of the act the subject performs upon an object”) is activated. The unaltered crab is in *Schutzstimmung* (“protective mood”); the de-shelled crab is in *Wohnstimmung* (“dwelling mood”); the starving crab is in *Fressstimmung* (“feeding mood”). The *Wirkbild* corresponding to the active *Stimmung* “completes and alters the perception image” (p. 97), and the resulting completed perception is *toned*: the anemone now has a *Schutzton* (“protection tone”), or a *Wohnton* (“dwelling tone”), or a *Fresston* (“feeding tone”). *The same external object is constituted as three different objects by three different tones, and the difference is generated by the subject’s mood, not by the object’s properties.*

3.2 The Doctrine of *Wirkton*: Pragmatic Disclosure of the Object

Uexküll generalizes this finding into a structural doctrine. The *Wirkton* is a perceptual quality that is *as* immediately given to the subject *as* shape, color, texture, or any other classical sensory quality. The chair is “sittable” with the same perceptual immediacy with which it is “wooden” or “brown.” The ladder is “climbable” with the same immediacy with which it is “tall” or “vertical.” The *Wirkton* is not a judgment added on top of a neutral perception; it is part of the perception itself, and indeed it is the part that organizes all the others into a single object. Uexküll’s example: “An African native, brought into a European house, sees neither chairs nor stairs nor ladders—only black bars and holes. Once shown how to climb the ladder, he sees it as climbable; the ladder has acquired its *Klettertönung* [‘climbing tone’]” (paraphrasing pp. 98–99). Before the *Wirkton* is acquired, there is no ladder—there are only bars and holes. With the *Wirkton*, there is a ladder.

The general formula Uexküll gives: “Every object that an animal encounters in its life is a thing of some functional kind—it has a sitting tone, a feeding tone, a flight tone, a swallowing tone, a climbing tone, etc. *Wirköne* are not added on top of an objective perception; they are constitutive of the perception itself. The object is not a “what” with “uses”; the object is a “for-this-action,” and that “for-which” is the *Wirkton*.” Uexküll lists, in passing through the chapter, an open-ended catalog: *Sitzton* (“sitting tone”), *Schutzton* (“protection tone”), *Wohnton* (“dwelling tone”), *Fresston* (“feeding tone”), *Hindernis-Ton* (“obstacle tone”), *Klettertönung* (“climbing tone”), *Tragtönung* (“carrying tone”), *Weg-ton* (“path tone”), *Wurf-ton* (“throwing tone”), *Schlagton* (“striking tone”; the chair-as-weapon case), *Nutztton* (“use tone”), *Gefahrton* (“danger tone”), and others. The catalog is intentionally open: there are as many *Wirköne* as there are functional cycles in which an object may figure for some subject.

3.3 The Reciprocal Specification of *Umweltsicherheit* (“environmental security”)

Uexküll draws from this analysis a counterintuitive corollary that is worth registering for its bearing on the philosophical treatment of *Umwelt*-richness. He observes that “the number of objects in a subject’s *Umwelt* equals the number of actions available to it” (p. 100), since each object is constituted by a *Wirkton*, and each *Wirkton* corresponds to an action-possibility. From this it follows that *the impoverishment of an Umwelt is identical with the simplification of its tonal palette*. The paramecium, with effectively one action (avoid obstacles), inhabits a world in which every object bears the single *Hindernis-Ton* (“obstacle tone”). The fly, with a slightly larger repertoire, sees most things under a *Lauf-tönung* (“running tone”)—the surface as that across which it can run—and only a few things differently. The dog has a richer palette (*Sitzton*, *Fresston*, *Hindernis-Ton*, etc.) but still less rich than the human’s. The human room contains the same physical objects as the fly room and the dog room, but the human room is richer because its tonal palette is richer; the fly’s room is the same room *otherwise tonal*.

This yields a thesis Uexküll calls the principle of *Umweltsicherheit* (“environmental security”): the simpler the *Umwelt*, the more secure the orientation. A paramecium cannot be uncertain whether to flee or to feed—every object bears the obstacle tone, and the response is unambiguous. A human, by contrast, lives in an *Umwelt* whose tonal palette is so rich that the same object may bear several *Wirk-töne* simultaneously (the chair as *Sitzton* and *Schlagton* both)—and the activated tone depends on *Stimmung*, on context, on the cultivation of the subject. *Sicherheit* trades off against *Reichtum* (“richness”). The tick’s existence is not “lower” than the human’s; it is one in which the tonal field has been reduced to absolute certainty.

3.4 The Dependence on *Stimmung*: Mood as the Organizer of Tones

The structural finding from the hermit-crab analysis is that the *Wirkton* depends on the *Stimmung* of the subject. *Stimmung* is the internal state—Uexküll uses the word with full force, in the same idiomatic register that Heidegger will later activate—that determines which of multiple possible *Wirkbilder* is ac-

tivated for a given *Wahrnehmungsbild*. The same object can carry multiple effect images (the chair as seat, as weapon, as obstacle, as decorative furniture); the subject's *Stimmung* picks one. "The *Stimmung* is the internal disposition that decides which effect image dominates and accordingly which tone the perception bears" (paraphrasing pp. 97, 99).

The structural homology with what Heidegger will name *Befindlichkeit* in *Sein und Zeit* §29 is, on the textual evidence, not coincidental. *Stimmung* names, in Uexküll's biology, what *Befindlichkeit* will name in Heidegger's existential analytic: the *prior, constitutive disposition that determines how things show up as significant before any explicit judgment, perception, or volition*. Heidegger reads *Stimmung* as the ontic manifestation of *Befindlichkeit*; Uexküll, ten years earlier, reads *Stimmung* as the dispositional ground that selects which *Wirkbild* will operate on the *Wahrnehmungsbild* and so which *Wirkton* the object will bear. The two analyses are formally identical, with one decisive difference: Heidegger insists on the *worldhood* of *Stimmung* (mood discloses *the world as a whole*), where Uexküll, working biologically, treats *Stimmung* as the internal state of an organic *Subjekt*. The relation between them is one of structural analogy with a residual ontological difference, and it is to this difference, finally, that the question of what kind of philosophy of life Uexküll's biology offers must be addressed.

4. *Suchbild* and *Suchton*: Function before Form (Foray Chapter 13)

4.1 The Search Image: Internal Templates that Override Perception

Chapter 13 returns to the tonal architecture from a different angle: the question of how the subject *finds* the objects whose *Wirkton* it would activate. Uexküll opens with two anecdotes whose force is phenomenological, not merely illustrative. *First*: a guest at his table, accustomed to a clay water-pitcher (*Wassertonkrug*), is presented with a glass carafe in the same place. The guest does not see the carafe—he asks for water and is told it is on the table; he searches the table; he does not see the carafe; only when his attention is drawn directly to it does the carafe become visible. The carafe was not occluded; it was *invisible*. The *Suchbild* ("search image") of "water pitcher" was geared to the clay pitcher's shape and color, and the carafe—departing from this *Suchbild*—failed to register. *Second*: a saleswoman cannot perceive a 100-mark banknote that is propped on edge in the till instead of lying flat; her *Suchbild* of "banknote" requires the bill to lie horizontally, and the upright bill is invisible to her even as her eyes pass directly over it. "The *Suchbild* extinguishes the *Wahrnehmungsbild*" (paraphrasing pp. 117–118).

The structural lesson is that perception, in any subject capable of search, is not stimulus-driven from below but template-driven from above. The *Suchbild* is an internally generated optical perception image that the subject projects outward into the *Umwelt*; it operates *before* any external stimulus and

organizes what counts as a perceivable object in the search. The dog hunting for its master's stick has a determinate *Suchbild* (this stick, of approximately this size and color); the toad, hungry, has a *Suchbild* of "worm-shape" or "spider-shape," depending on which it has just eaten and which it now expects.

4.2 The Search Tone: The Function-Tonal Mode

Uexküll then introduces the structural complement to *Suchbild*: *Suchton* ("search tone"). The distinction is decisive for the entire tonal architecture. A *Suchbild* searches for a *form*; a *Suchton* searches for a *function*. The dog with a determinate stick-image has a *Suchbild*; the human looking for "any seat" in a crowded waiting room has a *Suchton*—the perception is geared not to *this chair* but to *whatever can be sat on*. The toad, when it has just eaten, returns to a *Suchbild* (worm-form, spider-form). The hungry toad, before its first meal of the day, operates with a *Suchton*—an undifferentiated *Fresssuchton* ("feeding-search-tone") whose object is "anything edible." Only after its first prey is consumed does the *Suchton* sharpen into a determinate *Suchbild* of that particular prey-form.

Uexküll's clearest statement of the priority of *Suchton* over *Suchbild*: "The *Suchton* is more fundamental than the *Suchbild*; it is the function-orientation that makes form-search possible. The animal's first orientation in the world is *Suchton*; the *Suchbild* is a *Suchton* that has been crystallized around a recurrent particular form" (paraphrasing p. 121, with the hermit crab and toad cases). *Function precedes form*—a thesis that, at the biological level, accomplishes structurally what Heidegger will accomplish at the existential level when, in *Sein und Zeit* §15, he insists that the hammer is encountered first under the *Um-zu* ("in-order-to") of its readiness-to-hand, only secondarily as a present-at-hand thing with properties. The *Suchton* is the biological analogue of the *Um-zu*.

4.3 The Hermit Crab Revisited as Tonal Sequence

Uexküll uses Chapter 13 to recast the hermit-crab analysis of Chapter 9 in the language of *Suchton*. The hermit crab, in its three moods, is operating not with three *Suchbilder* but with three *Suchtöne*: *Schutzsuchton* ("protection-search-tone"), *Wohnsuchton* ("dwelling-search-tone"), *Fresssuchton* ("feeding-search-tone"). What it searches for is not "a sea anemone" but "anything that affords protection," "anything that affords shelter," "anything that affords nourishment." The sea anemone, in each case, *fits* the active *Suchton*—and the resulting completed perception is the tonal anemone (anemone-as-protection, anemone-as-shelter, anemone-as-food). Each of these three completed perceptions is what Uexküll, in Chapter 9, called the anemone-with-its-*Wirkton*; what Chapter 13 adds is that the *Wirkton* is selected by an antecedent *Suchton*, which is itself selected by the *Stimmung*.

The tonal architecture has now acquired a layered specificity. The *Stimmung* (mood, attunement) is the most general dispositional ground. Within the *Stimmung*, a *Suchton* (function-search-tone) is activated. The *Suchton* organizes the perceptual field so that a class of objects becomes salient. When such an

object is encountered, the *Suchton* crystallizes around its form into a *Suchbild* (form-search-image), and the *Suchbild* + *Wahrnehmungsbild* + *Wirkbild* combine to give the object its full *Wirkton* in the activated *Funktionskreis*.

4.4 *Leistungston*: The Function-Tone Itself

A further refinement that Uexküll introduces in Chapter 13 is the term *Leistungston* (“function tone,” “performance tone”). The German *Leistung* covers both “function” (what an object does) and “performance” (what it is achieved as doing); Uexküll uses the term to name the meaning-coloring an object has *qua functioning thing*, independent of its specific particular form. The chair has a *Sitzton*, but this is a *Leistungston*—any object that affords sitting bears it. The bowl has a *Trinkton* (“drinking tone”)—any object that affords drinking bears it. The *Leistungston* abstracts from the particular form to the function the form performs; it is the *function-tone* itself, whereas the *Wirkton* of a particular object is the function-tone *as embodied in this form*. The progression *Stimmung* → *Suchton* → *Leistungston* → *Wirkton* tracks the descent from the most general dispositional ground to the most particular tonal coloring of a specific object encountered.

5. *Magische Umwelten*: Free Productions of the Subject (*Foray Chapter 14*)

5.1 The Definition

Chapter 14, “Magical Environments,” is the most philosophically explosive chapter of the *Foray*. Uexküll defines a *magische Umwelt* (“magical environment”) as one that “contains very effective phenomena visible only to the subject, not bound to repeated experience” (p. 119); these are “free productions of the subject” (*freie Erzeugung des Subjekts*), perceptual phenomena that arise without any external stimulus correlate. The chapter contains four cases that bear directly on the tonal analysis.

First: the child playing Hansel and Gretel sees the witch “in the flesh.” Uexküll cites Frobenius’s *Paideuma*: a girl, playing the game, becomes terrified of an actual witch standing where no witch is. The *Wirkton* of “witchness” has been generated by the child’s *Stimmung*—the play-mood—and projected into the *Umwelt* without any external trigger.

Second: the dog perceives its master’s connection to its experiences “magically, not divided into cause and effect” (p. 121). The dog’s *Umwelt* is laden with *Tönungen* whose objective correlates are not isolable as causes; the master is grasped *as a tonal whole*, and the dog’s behavior follows from the tone, not from any chain of stimulus-and-response.

Third: the starling and the imaginary fly. Uexküll reports an experiment in which a starling, raised in captivity and never exposed to flies, performs the complete fly-catching behavioral sequence on an empty patch of air. “The environment is laden with feeding tone, which forces the perception image without sensory stimulus” (paraphrasing pp. 121–122). The starling has not learned to catch flies; the *Fresston* of its *Umwelt* generates a fly-shaped *Wahrnehmungsbild* without any external fly being present. The starling lives in an *Umwelt* in which fly-shaped objects can be perceived even when no fly-shaped object is there.

Fourth: the pea weevil larva, which bores an exit channel through still-tender pea flesh for the future use of the adult that has not yet emerged. The *Wirkmal* (the channel) is constituted by a future *Wirkton* that the larva itself does not yet experience; the *Umwelt* of the larva is laden, magically, with a *Tönung* that is not its own present mood but a *Bauplan*-tone of the species.

5.2 The Implication for the Tonal Doctrine

The structural significance of these cases is that *the tonal architecture is not exhausted by stimulus-driven perception*. *Töne* can be generated by the subject’s *Stimmung*, *Suchton*, or *Bauplan* (“body-plan”) in ways that produce *Wahrnehmungsbilder* without external correlate. The *Umwelt* is therefore not merely a receptive screen on which *Töne* are projected from external objects; it is a genuinely *generative* field in which the subject’s tonal apparatus *produces* perceptual phenomena. The *magische Umwelt* is the limit case of this generativity, where the tonal field operates without any stimulus correlate at all.

This has decisive consequences for the question of how *Töne* relate to the world. If *Töne* were merely passive registrations of object-properties, the *magische Umwelt* would be impossible. The *Töne* must be, in their primary function, *active dispositional structures of the subject*—the same structures that organize stimulus-driven perception—operating, in the magical case, *without* a stimulus to organize. The biological-philosophical conclusion is that *Töne* are constitutive of the *Umwelt* from the side of the *Subjekt*; they are *what the subject contributes* to the constitution of its world. The *Umwelt* is not the world *as it is* tinted by tones; the *Umwelt* is the world *as it is generated* by the subject’s tonal apparatus, with stimulus correlates supplying *only the occasions* on which the apparatus operates.

6. The Oak Paradigm: The Tonal Constitution of the *Umwelt* (Foray Chapter 15)

6.1 The Oak in Ten Environments

Chapter 15, “The Same Subject as Object in Different Environments,” gives the masterstroke of the *Foray*’s perspectivism. Uexküll takes a single oak tree and traces its appearance across ten *Umwelten*: forester, little girl, Russian seasonal workers, fox, owl, squirrel, songbird, ant, bark beetle, and

ichneumon wasp. Each *Umwelt* registers a different *Tönung* of the oak; in each, a different aspect of the oak is *cut out* by the active tonal articulation, and the rest is unregistered—not unseen, not occluded, but *not part of the Umwelt at all*. The oak in each *Umwelt* is, accordingly, a different object: the same physical tree, but ten different *Bedeutungsträger* (“carriers of meaning”).

The catalog (with the *Tönung* terms Uexküll uses where given, and structural reconstructions where not):

- **Forester:** the oak appears under the *Nutzton* (“use tone”). The trunk is registered as *cords of wood*; the rest of the tree is, for the forester, not relevant—not perceived under any *Wirkton*, hence not in the *Umwelt* in any robust sense. The forester’s *Wahrnehmungsbild* of the oak is “wood-mass-to-be-cut.”
- **Little girl:** the oak appears under a *Gefahrton* (“danger tone”) within a *magische Umwelt*. The bark’s gnarls and burls cohere into the face of a demon; the *Suchton* is “look-out-for-danger,” and the oak supplies the form. The girl’s *Wahrnehmungsbild* is “demon-with-wicked-face.”
- **Russian seasonal workers:** a fungus-growth on the oak appears under a *magischer Ton* (“magical tone”)—a religious tone within a peasant cosmology. The fungus is registered as a miraculous icon of the Virgin. The *Wahrnehmungsbild* is “Madonna in the wood.”
- **Fox:** the root cavity appears under a *Schutzton* (“protection tone”); the oak’s root system is registered as a protective roof under which the fox dens. The rest of the tree is not registered.
- **Owl:** the upper branches appear under a *Schutzton*; the oak’s high crown is a protective wall against ground predators. The *Wahrnehmungsbild* is “shelter from beneath.”
- **Squirrel:** the trunk and branches appear under a *Klettertönung* (“climbing tone”); the oak is registered as a vertical ladder of footholds. The *Wahrnehmungsbild* is “climbable structure.”
- **Songbirds:** the twigs appear under a *Tragtönung* (“carrying tone”); the oak’s twig-network supplies nest-supports. The *Wahrnehmungsbild* is “supports for the nest.”
- **Ants:** the bark surface appears under a hunting *Suchton*; the bark’s grooves are registered as paths along which prey may be encountered.
- **Bark beetle:** beneath-the-bark appears under both *Fresston* (“feeding tone”) and *Wohnton* (“dwelling tone”); the inner bark is food and the carved galleries are nursery.
- **Ichneumon wasp:** the wood appears under a penetration tone; the oak’s wood is registered as “soft-as-butter,” yielding to the wasp’s ovipositor.

6.2 The *Umwelt* as Tonal Cut

Uexküll's interpretive conclusion: each *Umwelt* "cuts out" of the oak just those features that can serve as *Merkmalträger* and *Wirkmalträger* for the active *Funktionskreis* of the inhabiting subject; the rest of the oak is *not in the Umwelt*—it is invisible, inaudible, intangible, *because the subject's tonal apparatus has no Funktionskreis in which it could figure*. "The oak-in-itself, considered apart from any *Umwelt*, is solidly put together—but it is *never known by all the subjects*, and *never knowable for them*" (paraphrasing p. 136). Summing all ten perspectival cuts does not yield the oak-in-itself; it yields only "chaos," because the *Umwelten* are incommensurable—each is its own complete world, and the cuts cannot be reassembled into a single object.

The relation between *Umwelt* and *Ton* is now visible in its full architecture. *The Umwelt is constituted as the tonal coloration of objects by the subject's active functional cycles*. Where there is no *Wirkton* for an object, there is no object in the *Umwelt*. The sum of the active *Wirkttöne* (and the *Suchttöne* and *Stimmungen* that select them) is the *Umwelt*. The *Umwelt* is not a container that *holds* tones; the *Umwelt* is the field that the tones *constitute*. Without tones there is no *Umwelt*; with tones, the *Umwelt* is *just* the field of tonal articulations of objects.

6.3 *Subjektive Wirklichkeit* and the Limits of Objectivism

Uexküll's foundational principle, stated in Chapter 15: "Each and every subject lives in a world in which there are only subjective realities [*subjektive Wirklichkeiten*]" (p. 130). The *Umwelt* contains only *subjektive Wirklichkeit*; the objective surroundings (*Umgebung*) never appear *as such* within any *Umwelt*. The principle is Kantian, but it is given biological articulation: the *Subjekt* of the *Umwelt* is not a transcendental ego but a living organism with a particular body, a particular tonal apparatus, a particular set of *Funktionskreise*. The "subjective" of *subjektive Wirklichkeit* is the *Subjekt-Umwelt* closure, not an inner mental field.

The principle has consequences for how the relation between *Tönen* and *Umwelt* is to be understood. *Töne* are not objective properties of objects that the subject more-or-less faithfully registers; they are *the modalities in which the Subjekt-Umwelt closure articulates itself*. The oak's *Klettertönung* is not an objective property of the oak that the squirrel happens to register; it is the way the oak figures *for the squirrel within the squirrel's Umwelt*. In a different sense, the same *Klettertönung* is also not "merely subjective"—it is not an illusion or a projection onto a really-non-climbable oak. The oak is climbable, in the only sense in which "climbable" has biological meaning: it figures in the *Funktionskreis* of climbing for a being whose *Bauplan* makes climbing possible. The *Klettertönung* is the structural fit between the squirrel's body and the oak's surface, articulated in the squirrel's *Umwelt*.

7. *Bedeutungsträger* and the Theory of Meaning (*Bedeutungslehre* Chapter 1)

7.1 The Stone, the Bowl, the Ladder

The 1940 *Bedeutungslehre* extends the tonal analysis of the *Foray* into a full theory of meaning. The opening chapter—“Carriers of Meaning” (“*Bedeutungsträger*”)—gives the most precise formulation of how *Töne* and *Bedeutung* relate. The chapter opens with a stone. Considered in itself, apart from any subject-relation, the stone is a “relationless object” (*beziehungsloses Objekt*)—a mere physical thing without rank or salience among its properties. Picked up by a man walking on a path, the stone becomes a *Wegstein* (“path-stone”): its *Wegton* (“path tone”) emerges. Hurlled by the same man at a barking dog, the *same stone, with no change in its physical properties*, becomes a *Wurfstein* (“throw-stone”): its *Wurfton* (“throwing tone”) emerges. The stone’s properties have not changed; what has changed is the *Funktionskreis* in which the stone has been entered. The same stone is two different *Bedeutungsträger* under two different *Wirköne*.

Uexküll generalizes the case across his examples. The same glass bowl is a window when its transparency leads (the *Durchsichtigkeitston*, “transparency tone”), a vase when its curvature leads (the *Behältnisston*, “container tone”). The same upright pole-and-rung structure is a ladder under the *Klettertönung* of climbing, a fence under a *Hindernis-Ton* of barring entry. In each case, the structural finding is that *the rank of properties (essential vs. accidental) is generated by the Wirkton, not by the object’s intrinsic properties*. The Scholastic distinction between *essentia* (“essence”) and *accidentia* (“accidents”) presupposes a *Bedeutungsträger*—a meaning-carrier—and ranks properties according to which is leading and which is supporting *for the meaning*. There are no objective *essentia*; *essentia* is a tonal category.

7.2 *Bedeutungsverbraucher*: The Subject as Consumer of Meaning

The structural complement to *Bedeutungsträger* is *Bedeutungsverbraucher* (“consumer of meaning”): the *Subjekt*’s body, considered as the structural counterpart that “consumes” the meaning the carrier provides. The *Komplement* (“complement”) of the meaning is the part of the body—the hand for the path-stone, the arm for the thrown stone, the leg for the climbed ladder, the seat-bones for the chair—that *fits* the carrier’s *Wirkmalträger*. The *Subjekt* and the *Bedeutungsträger* are jointly composed for each other: the path-stone *fits* the foot, and the foot *fits* the path-stone, *because the Funktionskreis of walking has shaped both into mutual fitness*. The fly is fly-shaped because the spider’s web is fly-shaped, and the spider’s web is fly-shaped because the fly’s *Bauplan* shapes its meaning. *Bedeutungsträger* and *Bedeutungsverbraucher* are not joined by accident but by *Komposition* (“composition”).

The *Wirkton* now finds its full biosemiotic specification: it is the meaning-function that the *Bedeutungsträger* bears for the *Bedeutungsverbraucher* under the active *Funktionskreis*. Every *Wirkton* presupposes the structural fit between carrier and consumer; every *Funktionskreis* enacts a specific *Wirkton*.

by closing the loop between the perception-organ and the effect-organ on the side of the *Subjekt*, the perception-mark-carrier and the effect-mark-carrier on the side of the object. The *Umwelt* is the totality of all such *Funktionskreise* in their tonal coloration of objects.

8. *Eigenton*, *Bildungsmelodie*, and the *Kompositionslehre* (*Bedeutungslehre* Chapters 7, 12)

8.1 *Eigenton*: The Self-Tone of the Living Cell

The 1940 *Bedeutungslehre* extends the tonal analysis from the *Funktionskreis* level down to the cellular level, where *Töne* acquire their most basic and most surprising articulation. Each living cell, on Uexküll's account, has its own *Eigenton* ("self-tone")—a specific energetic response to the stimulation it receives. The cell, in this sense, is a "bell" in the *lebendiges Glockenspiel* ("living carillon") of the organism: stimulated by an external factor, it sounds its *Eigenton*; the ensemble of *Eigentöne* across the cells of an organ produces an *Organmelodie* ("organ-melody"); the ensemble of *Organmelodien* across the organs of an organism produces an *Organismusymphonie* ("organism-symphony") or, in Uexküll's mature terminology, an *Urbild* ("primal image") of the organism.

The structural claim: the organism is not a machine of parts assembled by external forces but a *composition* whose parts are tones (*Eigentöne*) joined in melodies (*Organmelodien*) which join in a symphony (*Urbild*). The composition is governed not by mechanical causation but by a *Bedeutungsregel* ("meaning rule") that selects which environmental factors will become *Bedeutungsträger* for which cells and how the resulting tonal field will be organized. The *Bedeutungsregel* is the structural successor of the *Wirkton* at the cellular level: it is the rule that decides which stimuli are *for* which cells, just as the *Wirkton* decides which objects are *for* which actions.

8.2 *Bildungsmelodie*: The Formative Melody and Morphogenesis

A second level of tonal analysis applies to the developing organism. Uexküll proposes that morphogenesis—the formation of the organism from the germ cell—is governed by a *Bildungsmelodie* ("formative melody") that crystallizes out of the *Bedeutungskeim* ("meaning-germ"). The *Bedeutungskeim* is the originary meaning-nucleus from which the organism develops; the *Bildungsmelodie* is the unfolding of this meaning into the organic form. The acorn is the *Bedeutungskeim* of the oak; the egg is the *Bedeutungskeim* of the chicken. The genes are the *Klaviatur* ("clavier," "keyboard") on which the *Bildungsmelodie* plays; the protoplasm is the material the melody shapes.

The structural lesson is that *organic form is itself a tonal phenomenon*. The oak's particular shape is the *Bildungsmelodie* of its *Bedeutungskeim*; the squirrel's particular shape is the *Bildungsmelodie* of its *Bedeutungskeim*; and the structural fit between oak and squirrel that grounds the *Klettertönung* is itself

the result of two *Bildungsmelodien* that have been composed *contrapuntally* by Nature. The squirrel's body is shaped to climb the oak; the oak's bark is shaped to be climbed by the squirrel; both shapes are tonal articulations of a higher-order *Komposition*.

8.3 The *Kompositionslehre* and the *Partitur*

Uexküll's culminating thesis—stated with full hyperbole in Chapter 12 of the *Bedeutungslehre*—is the *Kompositionslehre der Natur* (“composition theory of nature”): Nature is the composer of all *Umwelten*, organisms are the instruments, *Wirköne* are the notes, and the whole is a *Partitur* (“score”) that, in the strongest formulation, is “beyond space and time” (*Bedeutungslehre* Chapter 12). The *Kompositionslehre* is Uexküll's response to the question of how the multiple *Umwelten* relate to each other and to Nature as a whole. The answer: they are *contrapuntal* parts of a single composition. The bee is composed for the flower; the flower is composed for the bee; the spider is composed for the fly; the fly is composed for the spider. Each *Funktionskreis* is a duet; the totality of all *Funktionskreise* is a symphony; the *Partitur* of the symphony is “Nature.”

The doctrine is, in its 1940 articulation, on the borderline of biology and metaphysics, and Uexküll is aware of the borderline. He insists that “Nature teaches no *Lehre* [‘lesson,’ ‘teaching’]”—the *Kompositionslehre* is generalized from observation, not deduced from principle. But the musical metaphor does substantial theoretical work: it makes intelligible how organisms can be jointly composed without a designer, how *Umwelten* can be incommensurable yet harmonious, how *Töne* can be both subject-side (the *Subjekt*'s tonal apparatus) and object-side (the *Bedeutungsträger*'s tonal coloration) at once. The *Komposition* is what holds the *Subjekt-Umwelt* closure together with the *Subjekt-Subjekt* attunement, all under the heading of a tonal architecture whose primary unit is the *Wirkton* and whose totality is the *Partitur*.

9. The Crossover with Aristotelian *Hedonic Tonality* and Heideggerian *Stimmung*

9.1 The Three Tonal Doctrines

The dissertational interest of Uexküll's tonal analysis lies in its triangulation with two other tonal doctrines: Aristotle's account of the *hēdonē-lupē* (“pleasure-pain”) tonality co-given with every act of perception, thought, and emotion (*De Anima* II.2, 413b 22 sqq.; *Nicomachean Ethics* X.1–5; *Rhetoric* II.1, 1378a 19–22), and Heidegger's account of *Stimmung* as the ontic manifestation of *Befindlichkeit* (*Sein und Zeit* §29; *Grundbegriffe der Metaphysik* §§16–17). All three doctrines articulate, in different idioms, the same structural insight: that there is a basic affective-tonal coloration that accompanies every act of perception, cognition, and action, and that this tonal coloration is *constitutive*, not added on top.

For Aristotle, the basic tonality is *hēdonē-lupē*—pleasure and pain—and it is co-given (*hepetai*) with every perceiving, thinking, considering, and emoting (*NE* X.4, 1175a 10 sqq.). It is not added to perception as a separate quale; it is the *toning-up or toning-down* of the very actuality of perception. The *Rhetoric*’s analysis of the higher-order emotions makes the same structural point at the level of the *pathē*: every emotion is “co-given with pain or pleasure” (*Rhet.* II.1, 1378a 21–22), and the pleasure-pain tonality is what makes the emotion *be* a specific way of being-disposed.

For Heidegger, the basic tonality is *Stimmung*, and it is the ontic manifestation of *Befindlichkeit*—the existential structure of *finding-oneself-disposed* that is co-original with understanding (*Verstehen*) and discourse (*Rede*) in the constitution of being-in-the-world. *Stimmung* is not a feeling added to a neutral perception; it is the prior dispositional ground that determines how things show up as significant—as threatening, as inviting, as boring, as mattering—before any explicit judgment, perception, or volition. *Befindlichkeit* discloses the world *as a whole*, in a tonality that pervades every encounter with every entity.

For Uexküll, the basic tonality is *Wirkton* (and its associated *Suchton*, *Stimmung*, *Eigenton*, *Bildungsmelodie*), and it is the constitutive tonality of every object in the *Umwelt*—the tonal articulation through which an object becomes *what it is for the subject* in the active *Funktionskreis*. The *Wirkton* is not an added layer; it is the structural fact that what the subject perceives is always perceived *under a function*, and the function colors the object before the object is fully constituted as an object.

9.2 The Structural Homology

The three doctrines are formally homologous in the following respects:

(1) **Co-giveness, not addition.** All three tonalities are *co-given* with the act they tone, not added on top of an antecedent neutral act. Aristotle’s *hēdonē* “follows” perception in the sense of *hepetai*—it is “with”-given, not consequent (Heidegger’s gloss in *GA* 18, p. 115). Heidegger’s *Stimmung* is co-original (*gleichursprünglich*) with *Verstehen* and *Rede* in the constitution of *Erschlossenheit* (“disclosedness”). Uexküll’s *Wirkton* is part of the *Wahrnehmungsbild* itself, not added by a subsequent judgment.

(2) **Constitutive of the object as object.** All three tonalities are *constitutive* of the object’s appearance *as* the object it is. Aristotle’s *hēdonic*-tonal articulation of the *pathē* makes anger *be* anger and fear *be* fear—the tonality is not something the *pathos* has but something the *pathos* is. Heidegger’s *Stimmung*-articulated entities show up *as* threatening or inviting; without the *Stimmung*, they do not show up *as* anything in particular. Uexküll’s *Wirkton*-articulated objects are the only objects in the *Umwelt*; without the *Wirkton*, the object is not in the *Umwelt* at all.

(3) **Subject-relative without being subjective in the bad sense.** All three tonalities are *Subjekt*-relative—they belong to the way a particular kind of subject relates to a particular kind of object—but none is “merely subjective” in the sense of an illusion or projection. Aristotle’s *hēdonē-lupē* are real determinations of the *psychē*’s being-in-its-world. Heidegger’s *Stimmung* discloses something about the world (the world *is* the way *Stimmung* discloses it, for Dasein in this *Stimmung*). Uexküll’s *Wirkton* is the real structural fit between subject and object in the *Funktionskreis*.

(4) **The basic dispositional axis of *hairesis* and *phugē*.** All three doctrines locate the basic tonal-dispositional axis in the *seizing* and *fleeing* of the subject toward and away from the toned object. Aristotle: pursuing the pleasant, fleeing the painful (*NE* II.3, 1104b 30; *EE* II.10, 1226b 10 sqq.; the Heideggerian gloss at GA 18, p. 166: “*hairesis* and *phugē* are the basic motivations of being-there”). Heidegger: the *Stimmung* of fear is a *recoiling-from*, the *Stimmung* of trust an *opening-toward*. Uexküll: the *Wirkton* of the obstacle is *avoid*; the *Wirkton* of the food is *seize*; the *Wirkton* of the predator is *flee*; the *Wirkton* of the mate is *approach*. The *hairesis-phugē* axis is the basic dispositional axis of all three tonal doctrines.

9.3 The Decisive Difference: World vs. *Umwelt*

The structural homology is real; but the doctrines diverge on a single decisive point that cannot be smoothed over. For Heidegger, *Stimmung* discloses *world* (*Welt*)—understood as the meaning-totality within which Dasein finds itself. For Uexküll, *Wirkton* constitutes *Umwelt*—understood as the closed perceptual-effective field of a particular biological subject. Heidegger’s *Welt* is not summable across Daseins (and not just because Daseins are particular); Uexküll’s *Umwelten* are at least in principle catalogable across species (the oak appears in the squirrel-*Umwelt*, the bark-beetle-*Umwelt*, etc., and the catalog is finite). Heidegger’s *Welt* has, in *Sein und Zeit*, a holistic-existential structure that cannot be biologically articulated; Uexküll’s *Umwelt* is biologically articulable in the architecture of *Funktionskreise*, *Bedeutungsträger*, *Bauplan*, *Bildungsmelodie*.

This difference matters for the question of *what kind of philosophy of life Uexküll’s tonal doctrine is*. It is not phenomenology in Heidegger’s sense, because the *Subjekt* whose *Umwelt* is articulated is not Dasein but a biological organism whose *Bauplan* is itself a structural condition for the kinds of *Wirk-töne* its *Funktionskreise* can register. It is not naive empiricism, because the *Wirkton* is not stimulus-driven but constitutive—the object is *for-an-action* before it is registered as an object. It is something else: a *biosemiotics* that sits between phenomenology and biology, articulating, in the language of *Töne* and *Funktionskreise*, what phenomenology articulates in the language of *Stimmung* and *Befindlichkeit*, and what Aristotelian *psyche*-analysis articulates in the language of *hēdonē-lupē* co-given with *aisthēsis*. The three vocabularies do not translate without remainder; but they are formally homologous in their tonal structure, and the homology is, on the textual evidence, not coincidental but reflective of a shared philosophical-biological problematic in the German-speaking inter-war period.

9.4 The Specific Gain for Aristotelian Hedonic Tonality

The bearing on the dissertational thesis—that Aristotle posits a basic *hedonic tonality* (good/bad, pleasure/pain, pursue/avoid) accompanying every moment of perception, thinking, considering, and emoting, and that this tonality is also a constitutive moment of the higher-order emotions of the *Rhetoric*—is this. Uexküll’s biology of *Töne* gives the *hedonic tonality* a specific structural articulation that is unavailable in the Aristotelian text alone. Where Aristotle’s *hēdonē-lupē* is asserted as co-given with *aisthēsis* but not given a structural specification of *how* it is co-given, Uexküll’s *Wirkton* gives the structural specification: the *hedonic-tonal* coloration is generated by the *Funktionskreis* in which the perceiver is engaged with the perceived; the active *Funktionskreis* is selected by the *Stimmung* of the subject; and the resulting *Wirkton* is the way the object figures *for* the active functional cycle. The *hedonic tonality* of Aristotelian perception is, in Uexküllian terms, the *Wirkton* that the active *Stimmung* and *Suchton* have selected for the *Funktionskreis* of the perceiving organism; and the *hedonic tonality* of the higher-order *pathē* of the *Rhetoric* is the *Wirkton* that the active social-evaluative *Funktionskreis* selects for the object of the *pathos* (the slighter-as-target-of-anger, the fear-inducer-as-threat, the pity-inducer-as-undeservedly-suffering).

The convergence is not a translation but an articulation. Aristotle says *what* the tonality does (it pursues, it flees, it makes pleasant or painful); Uexküll says *how* the tonality is generated (by the *Stimmung*-selected *Funktionskreis* operating through *Suchton* on the *Wahrnehmungsbild*); Heidegger says *what kind of being* requires such a tonality (*Dasein*, whose being is at issue in its world). The three doctrines, taken together, give a more complete account of the *hedonic-tonal* constitution of perception and emotion than any one of them gives alone. The dissertation that uses the term *hedonic tonality* to name the basic affective valence accompanying every act of *aisthēsis* and every articulation of *pathē* is on Aristotelian textual ground; on Uexküllian biosemiotic ground; on Heideggerian existential-analytic ground; and the convergence of the three is what gives the term its full philosophical weight.

10. Synthesis: Five Theses

Drawing the threads of the analysis together, Uexküll’s reading of the tonal architecture of the *Umwelt* in the *Foray* and the *Bedeutungslehre* issues in five master theses, each of which redirects the philosophical analysis of the living being away from the post-Cartesian, post-Darwinian biological framework.

Thesis I: The *Umwelt* is the closed self-world of a *Subjekt*, not a section of an objective world. The *Umwelt* (“surrounding-world,” “phenomenal world,” “self-world,” “environment”) is composed of *Merkwelt* (“perception world”) and *Wirkwelt* (“effect world”) joined in a closed *Funktionskreis* (“func-

tional cycle”). Each *Subjekt* has its *Umwelt*; no *Umwelt* is a part of any other; the *Umwelt* is not a portion of the *Umgebung* (“surroundings”) seen from a particular angle but a self-enclosed totality whose contents exist only within it.

Thesis II: The *Umwelt* is constituted by *Töne* (tones), and the *Töne* are dispositional articulations from the side of the *Subjekt*. The *Wirkton* (“effect tone”) is the meaning-coloration that an object bears under the active *Funktionskreis*; the *Suchton* (“search tone”) is the function-orientation that selects which class of objects can be encountered; the *Stimmung* (“mood,” “attunement”) is the dispositional ground that selects which *Suchton* is active. The tonal architecture descends from *Stimmung* through *Suchton* and *Leistungston* (“function tone”) to the *Wirkton* of the particular object. Without tones there is no *Umwelt*; with tones, the *Umwelt* is the field of tonal articulations.

Thesis III: The *Wirkton* is the structural fit between *Bedeutungsträger* (“carrier of meaning”) and *Bedeutungsverbraucher* (“consumer of meaning”) in the active *Funktionskreis*. The same physical object is constituted as different *Bedeutungsträger* by different *Wirkttöne*: the stone is path-stone (*Wegstein*) under *Wegton* and throw-stone (*Wurfstein*) under *Wurfton*; the oak is cordwood for the forester, demon for the child, protective roof for the fox, climbable structure for the squirrel, and food-and-nursery for the bark beetle. The rank of properties (*essentia* vs. *accidentia*) is generated by the *Wirkton*, not by the object’s intrinsic structure.

Thesis IV: *Töne* are not exhausted by stimulus-driven perception; they include *freie Erzeugung des Subjekts* (“free productions of the subject”) in *magische Umwelten* (“magical environments”). The starling catches the imaginary fly because its *Umwelt* is laden with *Fressston*; the child sees the witch because the play-*Stimmung* generates a witchness-*Wirkton*; the pea-weevil-larva bores the future-exit channel because its *Umwelt* is laden with a *Bauplan*-tone of the species. The tonal apparatus generates perceptual phenomena as well as registering them; the *Umwelt* is a generative field.

Thesis V: The tonal architecture extends from the *Eigenton* of the cell through the *Bildungsmelodie* of the organism to the *Kompositionslehre* of Nature, and the totality is a *Partitur* (“score”). Each cell has its self-tone; each organ has its *Organmelodie* (“organ-melody”); each organism has its *Urbild* (“primal image”) as the symphony of its organs; the totality of all *Umwelten*, joined contrapuntally, is the *Partitur* of Nature. The *Komposition* is not designed by an external composer but is the structural fit of *Bedeutungsträger* and *Bedeutungsverbraucher* across all *Funktionskreise*. The bee is composed for the flower, the flower for the bee, the spider for the fly, the fly for the spider; and the *Wirkton* is the audible articulation of this composition at the level of any particular *Subjekt-Umwelt* closure.

These five theses, taken together, constitute Uexküll’s tonal-biosemiotic account of the *Umwelt*. The *Umwelt* is closed (Thesis I); it is constituted by tones (Thesis II); the tones are the structural fit between *Subjekt* and *Bedeutungsträger* (Thesis III); the tonal apparatus is generative as well as receptive

(Thesis IV); the tonal architecture extends from the cell to Nature (Thesis V). The tonal architecture is what allows Uexküll's biology to articulate, in the language of *Töne* and *Funktionskreise*, what phenomenology articulates in the language of *Stimmung* and *Befindlichkeit*, and what Aristotelian psychology articulates in the language of *hēdonē-lupē* co-given with *aisthēsis*. The convergence is, on the textual evidence, not coincidental: Heidegger explicitly cites Uexküll in *Sein und Zeit* §§14–18, and the formal homology between Uexküllian *Stimmung* and Heideggerian *Befindlichkeit*—as well as between Uexküllian *Wirkton* and Aristotelian *hedonic tonality*—is, on the present analysis, a real philosophical convergence that rewards detailed reconstruction.

11. The Bearing on the Dissertational Thesis

For the dissertation that takes the term *hedonic tonality* to name the basic affective valence (good/bad, pleasure/pain, pursue/avoid) accompanying every act of perception, thinking, considering, and emoting—and that takes this same hedonic tonality to be a constitutive moment of the higher-order *pathē* of Aristotle's *Rhetoric*—Uexküll's tonal analysis offers four specific gains.

First, a structural specification of *what* the hedonic tonality *is*. Where Aristotle asserts that *hēdonē-lupē* is co-given with *aisthēsis* but does not specify the structural mechanism, Uexküll's *Wirkton* gives the mechanism: the hedonic tonality is the meaning-coloration the object bears for the active *Funktionskreis* of the perceiver, and this coloration is generated by the *Stimmung*-selected *Suchton* operating on the *Wahrnehmungsbild*.

Second, a structural specification of *how* the hedonic tonality is *Subjekt-relative without being merely subjective*. The *Wirkton* is the structural fit between *Bedeutungsträger* and *Bedeutungsverbraucher*; it is real, not illusory, but it is real *only within the Funktionskreis*. The hedonic tonality is, in the same way, real—it is a structural feature of the perceiver-perceived relation—but only within the active functional engagement of the perceiver.

Third, a structural specification of the *layered architecture* of the tonality. The descent from *Stimmung* through *Suchton* and *Leistungston* to *Wirkton* gives a precise architecture for the layered emergence of the hedonic-tonal coloration: the most general dispositional ground (*Stimmung*) selects a class of function-orientations (*Suchton*); the function-orientation selects a class of relevant objects (*Leistungston*); and the encounter with a particular object generates the specific *Wirkton*. The Aristotelian-Heideggerian reader of this dissertation can map *Stimmung* onto Heideggerian *Befindlichkeit* and Aristotelian *diathesis*, *Suchton* onto Heideggerian *Vorhabe* (“fore-having”) and Aristotelian *hexis*, *Leistungston* onto Aristotelian *dynamis*, and *Wirkton* onto the specific hedonic-tonal coloration of the perceived object in the active *pathos*.

Fourth, a specification of the *constitutive role of tonality in the higher-order emotions*. The *Wirkttöne* of *Schutzton*, *Fresston*, *Schlagton*, *Gefahrton*, *Liebestönung* (“love tone”), *Feindeston* (“enemy tone”) that Uexküll catalogs at the biological level have direct counterparts in the higher-order *pathē* of *Rhetoric II*: *Schutzton* is the tonal moment of *thrasos* (“confidence,” “courage”) in defended being-with-others; *Fresston*, in its sublimated rhetorical articulation, is the *epithumia* (“desire”) of acquisitive *pathē*; *Schlagton* is the *orexis antilypēseōs* (“desire for retaliation”) of *orgē* (“anger”); *Gefahrton* is the *λύπη τις ἢ παραχή* (“a kind of pain or disturbance”) of *phobos* (“fear”); *Liebestönung* is the *philia* of friendship; *Feindeston* is the *echthra* of enmity. The dissertation that takes the higher-order *pathē* of the *Rhetoric* to be tonal articulations of a basic hedonic-tonal architecture finds, in Uexküll’s biology, a structural analogue at the perceptual-evaluative level that supports the architectural claim across the perception-emotion gradient.

12. Bibliographic Coda

The texts on which this analysis principally relies, with their corresponding chapters and key German terms:

- **Foreword (pp. 41–43).** Anti-mechanist declaration; *Subjekt* vs. *Maschine* (“subject vs. machine”); *Merkzeug* / *Werkzeug* (“perception-tool” / “tool”); *Merkwelt* / *Wirkwelt* (“perception world” / “effect world”); soap-bubble metaphor.
- **Introduction (pp. 44–52).** The tick paradigm; *Funktionskreis* (“functional cycle”); *Merkmalträger* / *Wirkmalträger* (“perception-mark carrier” / “effect-mark carrier”); *Gegengefüge* (“counterstructure”); *Merkmal* / *Wirkmal* (“perception mark” / “effect mark”); *Merkzeichen* / *Wirkzeichen* (“perception sign” / “effect sign”); *Merkorgan* / *Wirkorgan* (“perception organ” / “effect organ”); *Maschinenführer* (“machine operator”); the principle “all subjects are inserted into their environments to the same degree of perfection”; *Moment* (“moment”); *Richtungsschritt* (“directional step”); the Kantian connection.
- **Chapter 9, “Perception Image and Effect Image” (pp. 92–97; main analysis pp. 96–101).** *Wahrnehmungsbild* (“perception image”); *Wirkbild* (“effect image”); *Wirkton* (“effect tone”); *Stimmung* (“mood,” “attunement,” “tuning”); the hermit-crab experiment; *Schutzton* (“protection tone”), *Wohnton* (“dwelling tone”), *Fresston* (“feeding tone”); the dog-and-chair experiment with *Sitzton* (“sitting tone”); the African-and-ladder anecdote with *Klettertönung* (“climbing tone”); *Hindernis-Ton* (“obstacle tone”); the principle of *Umweltsicherheit* (“environmental security”).
- **Chapter 13, “Search Image and Search Tone” (pp. 113–118; main analysis pp. 117–121).** *Suchbild* (“search image”); *Suchton* (“search tone”); *Leistungston* (“function tone”); *Leistung*

(“function,” “performance”); the carafe and the banknote anecdotes; the dog and the toad cases; the priority of function over form.

- **Chapter 14, “Magical Environments” (pp. 119–125).** *magische Umwelt* (“magical environment”); *freie Erzeugung des Subjekts* (“free production of the subject”); the child-and-witch anecdote; the dog’s “magical” connection to its master; the starling catching the imaginary fly; the pea weevil larva.
- **Chapter 15, “The Same Subject as Object in Different Environments” (pp. 126–132).** *Tönung* (“tonal coloration”); *Nutzton* (“use tone”); *Gefahrton* (“danger tone”); *Schutzton* (“protection tone”); *Klettertönung*, *Tragtönung* (“carrying tone”); the oak in ten *Umwelten*; *Merkmalträger / Wirkmalträger* (revisited); the principle “each and every subject lives in a world in which there are only subjective realities” (*subjektive Wirklichkeit*); *Natur* (capitalized) as concealed totality; the doctrine that summing all perspectival features yields chaos rather than the thing-in-itself.
- **Bedeutungslehre Chapter 1, “Carriers of Meaning” (pp. 139–145).** *Bedeutungsträger* (“carrier of meaning”); *Bedeutungsverbraucher* (“consumer of meaning”); *Wegton* (“path tone”) and *Wurfton* (“throwing tone”); the stone, the bowl, the ladder; the reinterpretation of Scholastic *essentia / accidentia*; *Wohnwelt* (“dwelling-world”); *Wohnhülle* (“dwelling-shell”); *Komplement* (“complement”); the flower-stem paradigm.
- **Bedeutungslehre Chapter 7, “The Composition Theory of Nature” (pp. 171–181).** *Kompositionslehre der Natur* (“composition theory of nature”); *Kontrapunkt* (“counterpoint”); *Bedeutungsregel* (“meaning rule”); *Bedeutungsfaktor* (“meaning factor”); *Bauplan* (“body-plan”); the bat-moth, octopus-seawater, and tick-mammal contrapuntal tables; *Bedeutungswechsel* (“meaning-switch”) in the brown ground beetle (friend → food after mating).
- **Bedeutungslehre Chapter 12, “Summary and Conclusion” (pp. 200–208).** *Eigenton* (“self-tone”); *lebendiges Glockenspiel* (“living carillon”); *Organmelodie* (“organ-melody”); *Organismusymphonie* (“organism-symphony”); *Urbild* (“primal image”); *Bildungsmelodie* (“formative melody”); *Bedeutungskeim* (“meaning-germ”); *Klaviatur* (“clavier”); *Partitur* (“score”); the house-and-garden allegory; the principle that “all the properties of things are nothing other than the perception signs imprinted upon them by the subject.”

The cross-pipeline references that bear on the dissertational thesis include: *Sein und Zeit* §§14–18 (Heidegger on *Umwelt* and worldhood, citing Uexküll); §29 (Heidegger on *Befindlichkeit* and *Stimmung*); §§15–18 (Heidegger on *Zuhandenheit* (“readiness-to-hand”) and the *Um-zu* (“in-order-to”), structurally homologous with Uexküll’s *Wirkton*); the 1929/30 *Grundbegriffe der Metaphysik* §§42–45 (Heidegger on *weltarm* (“world-poor”) animal, drawing on Uexküll); GA 18 §§16–21 (Heidegger on the *pathē* of *Rhetoric* II, where the *hedonic tonality* of the higher-order emotions is given its phe-

nomenological articulation); Aristotle, *De Anima* II.2, 413b 22 sqq.; II.3, 414b 4–5; III.7, 431a 8–12; *Nicomachean Ethics* X.1–5, 1172b–1175a (the *hēdonē-lupē* analysis); *Rhetoric* II.1, 1378a 19–22 (the three constitutive aspects of *pathos*).

Uexküll’s tonal analysis, in its mature 1934-1940 articulation, accordingly stands at a triangulation point where Aristotelian psychology, Heideggerian phenomenology, and biological *Umweltlehre* (“environment-doctrine”) meet on a shared problematic: the constitution of the perceived-world by the tonal-dispositional architecture of the perceiving subject. The reading developed here works *through* Uexküll’s biological text rather than around it, drawing out the tonal architecture that the text itself articulates and tracing the structural homology with the Aristotelian and Heideggerian doctrines of tonality. The *Foray* and the *Bedeutungslehre* are accordingly best read as a sustained articulation of an analysis whose structural insight is shared with Aristotle and Heidegger, whose biological articulation is unique to Uexküll, and whose convergence with the existential-phenomenological and the Aristotelian psychological idioms is, on the textual evidence, not coincidental but reflective of a shared philosophical-biological problematic in the German-speaking inter-war period.

End of report.